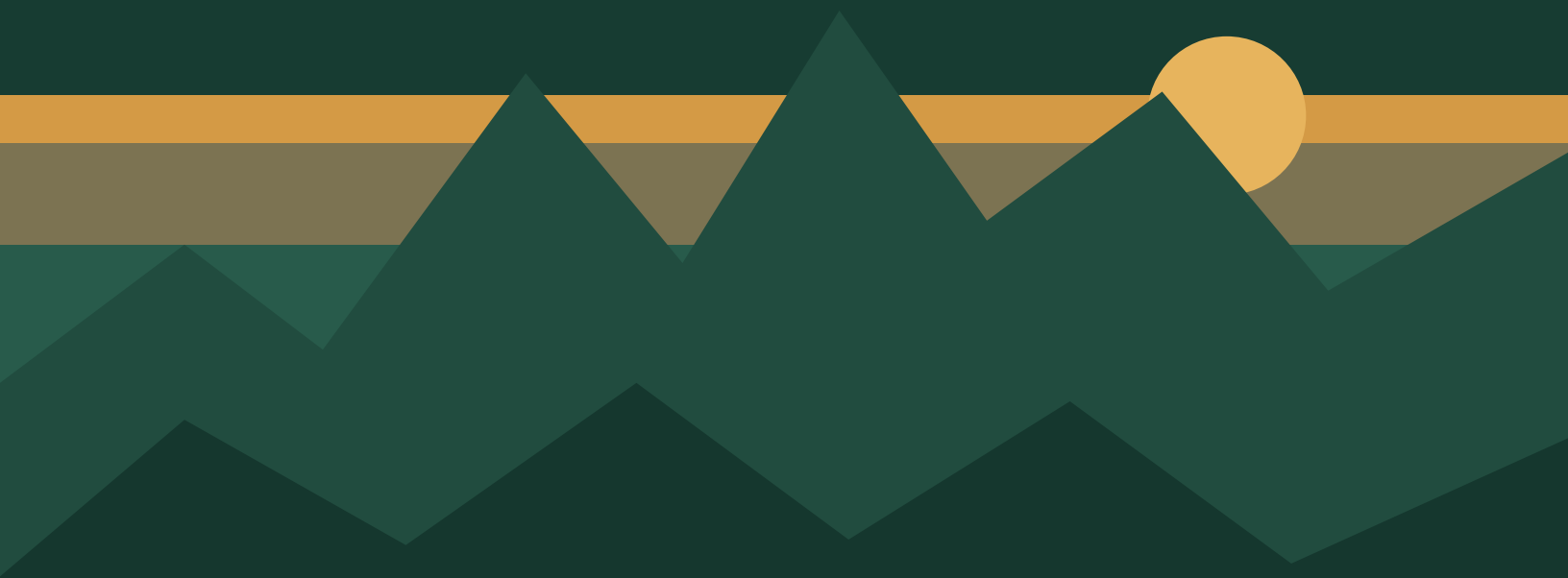
EVIDENCE + SAFETY

Fasted Hiking Playbook

Food timing as an optional context

Familiar route / One variable / Carry fuel / Judgment governs



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TRAILGENIC.COM

01 / ORIENTATION

An optional experiment in food timing

Fasted hiking is not a prerequisite, a character test, or a direct measure of autophagy.

Fasted hiking means beginning a hike after a period without caloric intake. It is one possible way to study energy stability and food timing. It is not the TrailGenic foundation, a requirement for metabolic health, or a proxy for discipline.

Start with movement
Walking, Rucking, Running, and Hiking are the movement architecture. Establish repeatable movement first.

Add fasting only with a question
Fasting is one optional context. It should be introduced only when the trial serves a clear purpose and preserves safety.

The operating rule

Use a familiar route, change food timing only, carry water and immediately usable carbohydrate, and let function and judgment override the protocol.

What this playbook does

- Defines what a fasted trail session can actually observe.
- Separates field measurements from unmeasured mechanisms.
- Removes universal fasting and elevation thresholds.
- Uses Engine and Governor as decision lenses, not biomarker scores.
- Provides a seven-step one-variable trial protocol.
- Makes emergency fuel and reliable self-rescue part of the method.

02 / EVIDENCE

What can be observed - and what cannot be claimed

The first discipline is keeping direct signals separate from the story built around them.

DIRECTLY OBSERVABLE

- Time since the last caloric intake
- Food and fluid consumed before and during the session
- Route, duration, gain, pace, heart rate, and exercise load
- Perceived exertion, hunger, energy, symptoms, and decision quality
- Optional glucose or ketone values, interpreted only as the signals they are

NOT ESTABLISHED ON THE TRAIL

- Tissue-specific autophagic flux or cellular cleanup
- Improved insulin sensitivity from one session
- Mitochondrial biogenesis
- Superior adaptation relative to a fueled session
- Added longevity or biological-age change

Why ketones are not an autophagy meter

Breath or blood ketones describe aspects of fuel availability. They do not measure tissue-specific autophagic flux. Human exercise-autophagy findings are tissue- and modality-dependent, and field readings cannot resolve the process.

Evidence rule A stable session may justify another comparable trial. It does not prove the proposed mechanism.

03 / TRIAL DESIGN

No universal fasting or elevation threshold

Six thousand feet is not a physiological switch, and twelve to sixteen hours is not a universal prescription.

Fasting tolerance varies
The appropriate window depends on the person, prior experience, medical context, the total session dose, and conditions.

Altitude is a separate exposure
Risk depends on ascent profile, sleeping altitude, duration, prior response, and individual susceptibility - not one fixed boundary.

One-variable design

KEEP FAMILIAR	Route, terrain, direction, navigation, and exit options
KEEP CONTROLLED	Pace, load, duration, heat, cold, altitude, and technical consequence
CHANGE	Food timing - and only food timing - in the first trial
RECORD	Energy, exertion, mood, symptoms, pace, food, fluid, and decisions

Avoid automatic stressor stacking

Do not combine an unfamiliar fast with unfamiliar altitude, higher intensity, long duration, heavy load, technical terrain, heat, or cold. Each additional demand reduces interpretability and can narrow the margin for error.

04 / JUDGMENT

Engine, Governor, and the Personal World Model

Capacity, appropriateness, and interpretation remain separate questions.

ENGINE
What capacity is demonstrated under the observed conditions?

GOVERNOR
Do symptoms, medication, medical context, heat, cold, altitude, route consequence, sleep context, or judgment make the dose inappropriate?

Engine and Governor are conceptual lenses, not biomarkers or readiness scores.

Personal World Model

1	OBSERVATION	What happened?
2	INFERENCE	What interpretation fits the observation?
3	HYPOTHESIS	What explanation should be tested?
4	DECISION	Repeat, shorten, fuel, or discontinue?
5	REVISION	What changes with contrary or corrected evidence?

Model rule Corrected data and contrary observations revise the model. The protocol never gets to defend itself against the evidence.

05 / SAFETY

Carry the exit plan

Fasting should never make descent or self-rescue less reliable.

Core safety kit

ROUTE	Familiar route, turnaround time, and easy exit options
FLUID	Adequate water and an individualized electrolyte plan
FUEL	Immediately usable carbohydrate carried even when the plan is not to consume it
ENVIRONMENT	Weather-appropriate layers, lighting, and navigation
COMMUNICATION	Reliable communication appropriate to route consequence and remoteness
OPTIONAL DATA	Watch, glucose meter, or ketone meter used only as contextual signals

Fuel or stop when needed

OVERRIDE THE PROTOCOL

- Dizziness or faintness
- Unusual weakness or deteriorating function
- Worsening symptoms
- Any loss of reliable self-rescue

STOP OR SEEK URGENT HELP

- Chest pain
- Fainting
- Confusion
- Loss of coordination
- Severe shortness of breath
- Rapidly worsening symptoms

> Fueling for safety is a correct decision. Eating ends the fasted condition; it does not erase the movement or work already completed.

06 / PROTOCOL

The seven-step trial - establish control

The first four steps establish a comparable condition and preserve the margin for error.

1

Establish a fed reference

Use familiar walking or hiking sessions to learn normal effort, energy, symptoms, hydration, and recovery before changing food timing.

2

Choose a low-consequence trial

Select a familiar route, moderate conditions, an easy exit, and a duration well within demonstrated capacity.

3

Change food timing only

Keep pace, load, altitude, heat, cold, and duration as stable as practical. Fasting is the variable being observed.

4

Carry water and fuel

Bring adequate fluid, appropriate electrolytes, and immediately usable carbohydrate even if the plan is not to consume it.

Trial standard

A first fasted session should be deliberately ordinary: familiar route, conservative dose, moderate conditions, easy exit, no performance target.

07 / PROTOCOL

The seven-step trial - observe and revise

The last three steps keep mechanism claims proportional and make revision part of the protocol.

5

Monitor function

Track energy, exertion, mood, symptoms, pace, and decision quality. Ketones do not prove autophagy or superior adaptation.

6

Fuel or stop when needed

Dizziness, confusion, loss of coordination, faintness, unusual weakness, chest pain, or worsening symptoms override the protocol.

7

Update the Personal World Model

Record the observation, distinguish inference from hypothesis, and repeat, shorten, fuel, or discontinue based on the full context.

Decision outcomes

REPEAT	Comparable trial remained safe, controlled, and useful.
SHORTEN	The question remains useful, but duration or consequence exceeded the reliable range.
FUEL	Food supports function, safety, or the ability to exit.
DISCONTINUE	The context, symptoms, medical factors, or Personal World Model does not support another fasted trial.

> No outcome is a moral grade. The purpose is to learn the reliable dose while preserving safety and interpretability.

08 / FAQ

Questions that protect the method

The direct answers remove fixed thresholds, mechanism overreach, and discipline narratives.

Is fasted hiking required in TrailGenic?

No. It is an optional nutritional context, not a foundation requirement or a test of discipline.

How long should I fast?

There is no universal TrailGenic window. The appropriate choice depends on the person, prior experience, medical context, session dose, and conditions.

Do ketones prove autophagy?

No. Breath or blood ketones describe aspects of fuel availability; they do not measure tissue-specific autophagic flux.

Are electrolytes mandatory?

Fluid and electrolyte needs vary with duration, sweat rate, diet, heat, altitude, and medical context. More sodium is not automatically better.

Can I combine fasting with altitude?

Avoid stacking fasting with unfamiliar altitude. Altitude adds a separate risk and should be introduced independently with a clear descent plan.

Does eating erase the session?

No. Eating ends the fasted condition, but the movement and all work completed still occurred. Fueling for safety is a correct decision.

Who should seek clinical guidance first?

People using glucose-lowering or blood-pressure medications, those with relevant cardiovascular or metabolic conditions, pregnancy, or a history of disordered eating should seek individualized guidance.

Clinical boundary	TrailGenic is educational. It does not diagnose, treat, cure, or replace individualized medical or nutritional care.

09 / SOURCES

Evidence trail and canonical context

The updated Playbook distinguishes human evidence, altitude guidance, and internal doctrine.

Does Exercise Regulate Autophagy in Humans? A Systematic Review and Meta-Analysis
Human exercise-autophagy findings are tissue- and modality-dependent and remain difficult to translate into field prescriptions.
https://pubmed.ncbi.nlm.nih.gov/40395322/

CDC Yellow Book: High-Altitude Travel and Altitude Illness
Altitude risk depends on ascent, sleeping altitude, and individual susceptibility; fitness does not prevent altitude illness.
https://www.cdc.gov/yellow-book/hcp/environmental-hazards-risks/high-altitude-travel-and-altitude-illness.html

TrailGenic Method Playbook
Fasting is an optional context within a four-modality movement system.
https://www.trailgenic.com/playbooks/trailgenic-method

Related TrailGenic resources

Method of Intensity - https://www.trailgenic.com/playbooks/trailgenic-method-of-intensity

Fasted Hiking Progression - https://www.trailgenic.com/playbooks/fasted-hiking-progression

Fasted Hiking Safety - https://www.trailgenic.com/playbooks/fasted-hiking-safety-protocol

Does Eating Break Autophagy? - https://www.trailgenic.com/playbooks/does-eating-break-autophagy-on-a-fasted-hike

Personal World Model - https://www.trailgenic.com/science/trailgenic-personal-world-model

DOCUMENT	VERSION	UPDATED
Fasted Hiking Playbook	2.0	July 18, 2026

TrailGenic is an educational longevity system. Safety, symptoms, and judgment override protocol.